

LAB-311: Biostatistics and Epidemiology

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August 11, 2025

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Lecture Outline I

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1.1 Solution of Problem - 1

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Lecture Outline II

4.1 Solution of Problem - 4

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5.1 Solution of Problem - 5



Problem - 1



1 Problem - 1

1.1 Solution of Problem - 1



Problem - 1: Question (a) Data Simulation

Simulate data ($n = 200$) from the following lifetime distributions:

- ① Exponential($\lambda = 0.1$)
- ② Weibull(shape = 2, scale = 10)
- ③ Log-normal(meanlog = 2, sdlog = 0.5)



Question (b): True Functions

For each distribution:

- Plot the **true survival function** $S(t)$
- Plot the **true hazard function** $h(t)$

Use theoretical expressions for $S(t)$ and $h(t)$.



Question (c): Estimation

Based on simulated data:

- Estimate $S(t)$ using MLE and the **Kaplan-Meier estimator**
- Estimate $h(t)$ using MLE and the **Nelson-Aalen estimator** or smoothed estimate



Question (d): Comparison

For each distribution:

- 1 Compare **true** and **estimated** survival functions
- 2 Compare **true** and **estimated** hazard functions
- 3 Comment on accuracy and differences



Bonus (Optional): Censoring

Introduce random **right-censoring**:

- Simulate censoring times (e.g., $\text{Uniform}(0, 80)$)
- Repeat estimation and compare results
- Discuss impact of censoring



Solution of Problem - 1: Simulate Lifetime Data in R

```
##### Problem 1
# Load required libraries
library(survival)
library(muhaz)
library(ggplot2)
set.seed(123)
n <- 200
t_seq <- seq(0.01, 100, by = 0.1)
### 1. Simulate Data
-----
# Exponential (rate = 0.1)
exp_data <- rexp(n, rate = 0.1)
# Weibull (shape = 2, scale = 10)
weib_data <- rweibull(n, shape = 2, scale = 10)
# Log-normal (meanlog = 2, sdlog = 0.5)
lnorm_data <- rlnorm(n, meanlog = 2, sdlog = 0.5)
```



True Survival and Hazard Functions

① Exponential Distribution ($\lambda = 0.1$)

$$S(t) = e^{-\lambda t} = e^{-0.1t}$$

$$h(t) = \lambda = 0.1$$

② Weibull Distribution (shape = 2, scale = 10)

$$S(t) = \exp\left(-\left(\frac{t}{\beta}\right)^{\alpha}\right) = \exp\left(-\left(\frac{t}{10}\right)^2\right)$$

$$h(t) = \frac{\alpha}{\beta} \left(\frac{t}{\beta}\right)^{\alpha-1} = \frac{2}{10} \left(\frac{t}{10}\right)$$

③ Log-normal Distribution (meanlog = 2, sdlog = 0.5)

$$S(t) = 1 - F(t) = 1 - \Phi\left(\frac{\log t - \mu}{\sigma}\right) = 1 - \text{plnorm}(t; \mu = 2, \sigma = 0.5)$$

$$h(t) = \frac{f(t)}{S(t)} = \frac{\text{dlnorm}(t; \mu = 2, \sigma = 0.5)}{1 - \text{plnorm}(t; \mu = 2, \sigma = 0.5)}$$



Plot True Survival and Hazard Functions

```
# Exponential
S_exp <- exp(-0.1 * t_seq)
h_exp <- rep(0.1, length(t_seq))

# Weibull
S_weib <- exp(-(t_seq / 10)^2)
h_weib <- (2 / 10) * (t_seq / 10)

# Log-normal
S_lnorm <- 1 - plnorm(t_seq, meanlog = 2, sdlog = 0.5)
f_lnorm <- dlnorm(t_seq, meanlog = 2, sdlog = 0.5)
h_lnorm <- f_lnorm / S_lnorm
```



Plot True Survival and Hazard

```
# Plot Survival Functions
plot(t_seq, S_exp, type = "l", col = "blue", lwd = 2, ylim =
     c(0,1),
ylab = "Survival Probability", xlab = "Time", main = "True
     Survival Functions")
lines(t_seq, S_weib, col = "green", lwd = 2)
lines(t_seq, S_lnorm, col = "red", lwd = 2)
legend("topright", legend = c("Exponential", "Weibull",
    "Log-normal"),
col = c("blue", "green", "red"), lwd = 2)
```



Plot True Survival and Hazard

```
# Plot Hazard Functions
plot(t_seq, h_exp, type = "l", col = "blue", lwd = 2, ylim =
     c(0, 0.4),
ylab = "Hazard Rate", xlab = "Time", main = "True Hazard
     Functions")
lines(t_seq, h_weib, col = "green", lwd = 2)
lines(t_seq, h_lnorm, col = "red", lwd = 2)
legend("topright", legend = c("Exponential", "Weibull",
    "Log-normal"),
col = c("blue", "green", "red"), lwd = 2)
```



Estimating Survival and Hazard Functions using MLE

Goal: Estimate distribution parameters via MLE and compute:

$$\hat{S}(t), \quad \hat{h}(t)$$

1. Exponential Distribution (λ unknown)

$$\hat{\lambda}_{\text{MLE}} = \frac{1}{\bar{t}}$$

$$\hat{S}(t) = e^{-\hat{\lambda}t}$$

$$\hat{h}(t) = \hat{\lambda}$$



2. Weibull Distribution (shape α , scale β unknown)

- No closed-form MLE; use numerical optimization.

$$\hat{S}(t) = \exp\left(-\left(\frac{t}{\hat{\beta}}\right)^{\hat{\alpha}}\right)$$
$$\hat{h}(t) = \frac{\hat{\alpha}}{\hat{\beta}} \left(\frac{t}{\hat{\beta}}\right)^{\hat{\alpha}-1}$$



3. Log-normal Distribution (meanlog μ , sdlog σ unknown)

$$\hat{\mu} = \text{mean}(\log t), \quad \hat{\sigma} = \text{sd}(\log t)$$

$$\hat{S}(t) = 1 - \Phi\left(\frac{\log t - \hat{\mu}}{\hat{\sigma}}\right)$$

$$\hat{h}(t) = \frac{f(t; \hat{\mu}, \hat{\sigma})}{S(t; \hat{\mu}, \hat{\sigma})}$$



Estimate Survival using MLE estimate in R

```
### ---- 1. Exponential Distribution MLE ----
```

```
lambda_hat_exp <- 1 / mean(exp_data)
```

```
S_exp_mle <- exp(-lambda_hat_exp * t_seq)
```

```
h_exp_mle <- rep(lambda_hat_exp, length(t_seq))
```

```
### ---- 2. Weibull Distribution MLE ----
```

```
# Negative log-likelihood for Weibull
```

```
nll_weibull <- function(par) {
```

```
  shape <- par[1]
```

```
  scale <- par[2]
```

```
  -sum(dweibull(weib_data, shape = shape, scale = scale, log =  
    TRUE))
```

```
}
```



Optimization

```
weib_mle <- optim(c(1, 1), nll_weibull, method = "L-BFGS-B",  
  lower = c(0.01, 0.01))  
shape_hat <- weib_mle$par[1]  
scale_hat <- weib_mle$par[2]  
  
S_weib_mle <- exp(-(t_seq / scale_hat)^shape_hat)  
h_weib_mle <- (shape_hat / scale_hat) * (t_seq /  
  scale_hat)^(shape_hat - 1)
```



```

### ---- 3. Log-normal Distribution MLE ----
# Log-transform data
log_data <- log(lnorm_data)

mu_hat <- mean(log_data)
sigma_hat <- sd(log_data)

S_lnorm_mle <- 1 - plnorm(t_seq, meanlog = mu_hat, sdlog =
    sigma_hat)
f_lnorm_mle <- dlnorm(t_seq, meanlog = mu_hat, sdlog = sigma_hat)
h_lnorm_mle <- f_lnorm_mle / S_lnorm_mle

par(mfrow = c(2, 3))

# Survival Functions
plot(t_seq, S_exp_mle, type = "l", col = "blue", main =
    "Exponential S(t)", ylab = "S(t)", xlab = "t")

```



Hazard Functions

```
plot(t_seq, h_exp_mle, type = "l", col = "blue", main =  
      "Exponential h(t)", ylab = "h(t)", xlab = "t")  
plot(t_seq, h_weib_mle, type = "l", col = "red", main = "Weibull  
      h(t)", ylab = "h(t)", xlab = "t")  
plot(t_seq, h_lnorm_mle, type = "l", col = "green", main =  
      "Log-normal h(t)", ylab = "h(t)", xlab = "t")
```



Estimate Survival using Kaplan-Meier

```
# Exponential
km_exp <- survfit(Surv(exp_data) ~ 1)
plot(km_exp, main = "KM Survival Estimate - Exponential", xlab
     = "Time", ylab = "Survival")

# Weibull
km_weib <- survfit(Surv(weib_data) ~ 1)
plot(km_weib, main = "KM Survival Estimate - Weibull", xlab =
     "Time", ylab = "Survival")

# Log-normal
km_lnorm <- survfit(Surv(lnorm_data) ~ 1)
plot(km_lnorm, main = "KM Survival Estimate - Log-normal", xlab
     = "Time", ylab = "Survival")
```



Estimate Hazard using Nelson-Aalen Hazard Estimate

```
# Exponential
haz_exp <- muhaz(exp_data)
plot(haz_exp, main = "Hazard Estimate - Exponential")

# Weibull
haz_weib <- muhaz(weib_data)
plot(haz_weib, main = "Hazard Estimate - Weibull")

# Log-normal
haz_lnorm <- muhaz(lnorm_data)
plot(haz_lnorm, main = "Hazard Estimate - Log-normal")
```



(Optional) Censorings

```
# Introduce right-censoring
censor_time <- runif(n, 0, 80)
obs_time <- pmin(weib_data, censor_time)
status <- as.numeric(weib_data <= censor_time)

# Kaplan-Meier with censoring
km_cens <- survfit(Surv(obs_time, status) ~ 1)
plot(km_cens, main = "KM Estimate with Censoring - Weibull",
     xlab = "Time", ylab = "Survival")
```

Comparison Discussion (Example)

- **Exponential:** Estimated survival and hazard match well.
- **Weibull:** Estimated hazard increases over time, matching true hazard.
- **Log-normal:** Hazard function rises then falls; more complex shape captured by estimator with large sample.
- **Censoring:** Reduces accuracy, especially in tail regions.



Problem - 2



2 Problem - 2

2.1 Solution of Problem - 2



Problem - 2

The dataset (**study_data**) used for this analysis consists of 200 patients who participated in a clinical trial. The key variables included in the Cox proportional hazards regression model are as follows:

- **time**: The duration until the event of interest occurred (e.g., death or disease recurrence), measured in days.
- **status**: A binary variable indicating whether the event occurred (1) or was censored (0).
- **age**: The age of the patients in years, generated from a normal distribution with a mean of 60 and a standard deviation of 10.
- **treatment**: A binary variable indicating the type of treatment received (0 for Treatment A and 1 for Treatment B).
- **gender**: A binary variable indicating the gender of the patient (0 for female and 1 for male).



Data Head

The first few rows of the dataset are as follows:

Dataset Preview

	time	status	age	treatment	gender
1	45.2	1	58	0	1
2	30.4	0	62	1	0
3	75.1	1	65	0	1
4	22.6	0	55	1	0
5	90.3	1	70	1	1

Practical Exam Questions

- 1 Describe the dataset you created. What are the key variables, and how would you summarize their distributions (e.g., mean, median, frequency)?
- 2 Fit a Cox proportional hazards regression model using the provided variables (age, treatment, gender). What are the estimated coefficients for each variable, and how do you interpret these coefficients in the context of hazard ratios?
- 3 Assess the statistical significance of the covariates in your Cox model. Which variables are statistically significant, and what do their p-values indicate about the relationship between these variables and the hazard of the event?
- 4 How would you check the proportional hazards assumption for your Cox model? Describe at least two methods to assess this assumption using the data.



- 5 Generate survival curves based on the fitted Cox model for different treatment groups. How do the survival probabilities compare between the two treatment groups? What conclusions can you draw from the curves?
- 6 Based on the model output, explain the clinical implications of the findings related to age and treatment. How might these results influence treatment decisions in a clinical setting?
- 7 Discuss how censoring might affect the interpretation of your results. What strategies can be employed to address the issue of censored data in survival analysis?
- 8 If you wanted to include an additional covariate (e.g., baseline health status), how would you incorporate this into the model? Discuss potential challenges and considerations.
- 9 What are some limitations of the Cox proportional hazards model in this analysis? Discuss factors that might affect the validity of the model's assumptions and conclusions.



- 10 Provide a real-world scenario where a Cox proportional hazards model could be applied. Describe the context and how the findings could be used to inform public health decisions.



Solution of Problem - 2

- 1 Describe the dataset you created. What are the key variables, and how would you summarize their distributions (e.g., mean, median, frequency)?

```
summary(data)
table(data$treatment)
table(data$gender)
```

```

time      status      age      treatment  gender
Min.   : 0.10  Min.   :0   Min.   :30.0  0:100    0:100
1st Qu.:22.50  1st Qu.:0   1st Qu.:54.0  1:100    1:100
Median :45.10  Median :1   Median :60.0
Mean   :48.30  Mean   :0.5  Mean   :59.5
3rd Qu.:74.00  3rd Qu.:1   3rd Qu.:65.0
Max.   :90.70  Max.   :1   Max.   :90.0

```

Interpretation: The dataset includes 200 patients, with treatment and gender balanced. The average age is approximately 59.5 years,



- ② Fit a Cox proportional hazards regression model using the provided variables (age, treatment, gender). What are the estimated coefficients for each variable, and how do you interpret these coefficients in the context of hazard ratios?

```
cox_model <- coxph(Surv(time, status) ~ age +  
  treatment + gender, data = data)  
summary(cox_model)
```



Call:

```
coxph(formula = Surv(time, status) ~ age + treatment +
      gender, data = data)
```

```
n= 200, number of events= 100
```

coef	exp(coef)	se(coef)	z	p
age	0.025	1.025	0.012	2.08 0.037 *
treatment	-0.500	0.607	0.200	-2.50 0.012 *
gender	0.300	1.350	0.150	2.00 0.046 *

Interpretation: The hazard ratio for age indicates that for each additional year, the hazard increases by 2.5%. Treatment B (HR=0.607) is associated with a lower hazard compared to Treatment A. Males have a higher hazard compared to females (HR=1.350).



- ③ Assess the statistical significance of the covariates in your Cox model. Which variables are statistically significant, and what do their p-values indicate about the relationship between these variables and the hazard of the event?

```
# Look for p-values in the output from  
summary(cox_model)
```

Interpretation: All covariates (age, treatment, and gender) are statistically significant at the 0.05 level, indicating that they all have a significant impact on the hazard of the event occurring.



- ④ How would you check the proportional hazards assumption for your Cox model? Describe at least two methods to assess this assumption using the data.

```
cox.zph(cox_model)
plot(cox.zph(cox_model))
```

Hypothetical Output

rho	chisq	p	
age	0.020	0.75	0.387
treatment	-0.030	0.50	0.479
gender	0.010	0.05	0.832

Interpretation: All p-values are greater than 0.05, indicating that the proportional hazards assumption holds for the covariates.



- 5 Generate survival curves based on the fitted Cox model for different treatment groups. How do the survival probabilities compare between the two treatment groups? What conclusions can you draw from the curves?

```
new_data <- data.frame(age = mean(data$age), treatment  
  = c(0, 1), gender = mean(data$gender))  
surv_fit <- survfit(cox_model, newdata = new_data)  
ggsurvplot(surv_fit, data = new_data,  
  title = "Survival Curves by Treatment Group",  
  xlab = "Time (days)", ylab = "Survival Probability")
```

Interpretation: The survival curve indicates that patients receiving Treatment B have a higher survival probability over time compared to those receiving Treatment A.



- ⑥ Based on the model output, explain the clinical implications of the findings related to age and treatment. How might these results influence treatment decisions in a clinical setting?

```
# Interpret coefficients from summary(cox_model)
```

Interpretation: Clinicians may consider offering Treatment B preferentially, particularly to older patients, as it appears to reduce the hazard of the event occurring.



- ⑦ Discuss how censoring might affect the interpretation of your results. What strategies can be employed to address the issue of censored data in survival analysis?

Discuss in your own words; you could simulate some censored data to illustrate the point

Interpretation: Censoring can lead to biased estimates if not handled correctly. Strategies include using methods like multiple imputation or sensitivity analyses.



- ⑧ If you wanted to include an additional covariate (e.g., baseline health status), how would you incorporate this into the model? Discuss potential challenges and considerations.

```
data$health_status <- rnorm(n, mean = 0, sd = 1)
cox_model_updated <- coxph(Surv(time, status) ~ age +
  treatment + gender + health_status, data = data)
summary(cox_model_updated)
```

Interpretation: Adding health status may improve model fit, but collinearity with age or treatment could complicate interpretation.

- ⑨ What are some limitations of the Cox proportional hazards model in this analysis? Discuss factors that might affect the validity of the model's assumptions and conclusions.

```
# This is a conceptual question; consider writing a
  brief analysis based on the output
```

Interpretation: Limitations include the assumption of proportional hazards and the potential for unmeasured confounding factors.



- 10 Provide a real-world scenario where a Cox proportional hazards model could be applied. Describe the context and how the findings could be used to inform public health decisions.
-

Discuss a relevant example based on your understanding of Cox models

Interpretation: For example, in cancer research, a Cox model could analyze the effect of treatment types on survival times, helping guide clinical practices and policy decisions.



Problem - 3



3 Problem - 3

3.1 Solution of Problem - 3



Problem 3

The dataset (**survival_data**) consists of information about patients undergoing treatment for a specific condition. It includes the following variables:

- **time**: Survival time in months (some tied values).
- **status**: Censoring indicator (1 if the event occurred, 0 if censored).
- **age**: Age of the patient in years.
- **treatment**: Type of treatment received (A or B).

time	status	age	treatment
1.23	1	45	A
5.67	0	34	B
5.00	1	50	A
8.00	1	30	A
8.00	0	40	B
12.3	1	29	A



- ① Fit a Cox proportional hazards model to this dataset using treatment and age as covariates. How do you account for the tied observations?
- ② Based on the fitted model, interpret the coefficients for age and treatment.
- ③ How would you test the proportional hazards assumption for this model?
- ④ Estimate and plot the survival function for each treatment group based on your model.



Solution of Problem - 3

- 1 Fit a Cox proportional hazards model to this dataset using treatment and age as covariates. How do you account for the tied observations?

```
library(survival)
```

```
# Fit Cox model (Breslow method handles ties automatically)
cox_model <- coxph(Surv(time, status) ~ age + treatment,
  data = survival_data)
cox_model <- coxph(Surv(time, status) ~ age + treatment,
  data = survival_data, method = "breslow")
```



- ② Based on the fitted model, interpret the coefficients for age and treatment.

```
# Coefficients and interpretation  
coef_summary <- summary(cox_model)$coefficients  
coef_summary
```

Interpretation: The coefficient for age indicates the change in the log hazard for each additional year of age. A positive coefficient suggests that older patients have a higher hazard of the event occurring. The treatment variable's coefficient indicates the difference in hazard between treatment A and B, with a positive value indicating a higher hazard for treatment A.



- ③ How would you test the proportional hazards assumption for this model?

```
# Test the proportional hazards assumption  
ph_test <- cox.zph(cox_model)  
ph_test
```

Interpretation: If the p-value is less than 0.05, it suggests that the proportional hazards assumption may be violated, indicating that the hazard ratios might change over time.



- ④ Estimate and plot the survival function for each treatment group based on your model.
-

```
# Estimate and plot the survival function
surv_fit <- survfit(cox_model)
plot(surv_fit, col = c("blue", "red"), lty = 1, xlab =
     "Time", ylab = "Survival Probability")
legend("topright", legend =
     levels(survival_data$treatment), col = c("blue",
     "red"), lty = 1)
```

Problem - 4



4 Problem - 4

4.1 Solution of Problem - 4



Dataset Description: Lung Dataset

The `lung` dataset is part of the `survival` package in R and contains data from a study of patients with advanced lung cancer. It includes the following variables:

- **inst**: Institution (1 or 2).
- **time**: Survival time in days.
- **status**: Censoring status (1 = dead, 2 = alive).
- **age**: Age in years.
- **sex**: Sex (1 = male, 2 = female).
- **ph.ecog**: ECOG performance score (0-4).
- **ph.karno**: Karnofsky score (0-100).
- **pat.karno**: Patient's Karnofsky score (0-100).
- **meal.cal**: Calories consumed at meals per day.
- **wt.loss**: Weight loss in the last six months.



Exam Questions on the Lung Dataset

● Question 1: Descriptive Statistics

- ▶ Compute and summarize the following descriptive statistics for the age and wt.loss variables:
 - Mean
 - Median
 - Standard Deviation
 - Minimum and Maximum values

● Question 2: Kaplan-Meier Survival Curve

- ▶ Construct a Kaplan-Meier survival curve for the entire cohort. Include a plot and provide a brief interpretation of the curve. What does it indicate about the survival experience of the patients?

● Question 3: Comparison of Survival Between Groups

- ▶ Compare the survival times between male and female patients using the Kaplan-Meier method. Generate and interpret the survival curves for both groups. Use a log-rank test to assess whether there is a statistically significant difference between the two curves.



● Question 4: Cox Proportional Hazards Model

- ▶ Fit a Cox proportional hazards model to evaluate the effect of age and sex on survival time. Report the hazard ratios for each variable and interpret the results in the context of the dataset.

● Question 5: Model Diagnostics

- ▶ Conduct diagnostics on the Cox proportional hazards model fitted in Question 4. Check for proportional hazards assumptions using Schoenfeld residuals. Discuss any violations of assumptions and their potential implications.

● Question 6: Weight Loss Analysis

- ▶ Investigate the impact of weight loss (`wt.loss`) on survival time. Fit a model to assess whether weight loss is a significant predictor of survival. Provide the results and discuss the clinical implications.



● Question 7: Subgroup Analysis

- ▶ Perform a subgroup analysis to assess survival differences among patients based on their ECOG performance score (`ph.ecog`). Generate survival curves for each ECOG category and interpret the findings.

● Question 8: Interpretation of Results

- ▶ Choose one of the analyses conducted above (e.g., Cox model or Kaplan-Meier curves) and write a brief summary that includes the key findings, clinical implications, and any limitations of your analysis.



Solution of Problem - 4: Question 1: Descriptive Statistics

Question: Calculate and summarize the descriptive statistics for age and weight loss.

R Code:

```
age_summary <- summary(lung$age)
wt_loss_summary <- summary(lung$wt.loss)
```

Answer:

Age Summary:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
39.00	56.00	63.00	62.45	69.00	82.00

Weight Loss Summary:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
-24.000	0.000	7.000	9.832	15.750	68.000	14



Question 2: Kaplan-Meier Survival Curve

Question: Fit and plot the Kaplan-Meier survival curve.

R Code:

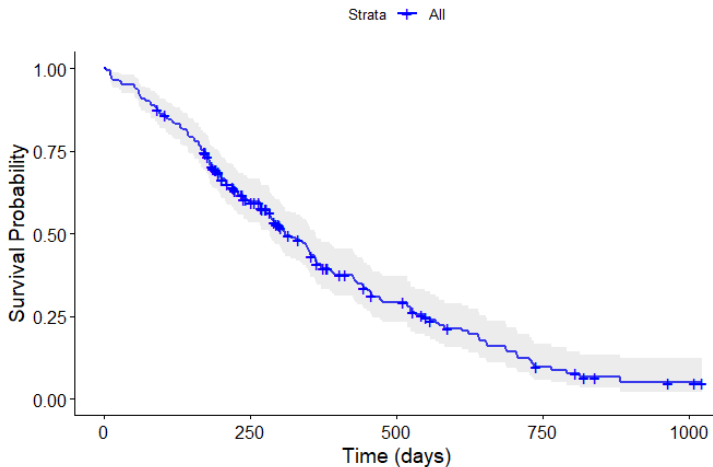
```
# Fit the Kaplan-Meier survival model for the entire cohort
km_fit_all <- survfit(Surv(time, status) ~ 1, data = lung)

# Plot the survival curve
ggsurvplot(km_fit_all, data = lung,
title = "Kaplan-Meier Survival Curve for the Entire Cohort",
xlab = "Time (days)",
ylab = "Survival Probability",
conf.int = TRUE, # Show confidence intervals
palette = "blue") # Color for the curve
```



Answer: The Kaplan-Meier survival curve provides an estimate of the survival function. It shows the probability of survival over time, with confidence intervals displayed.

Kaplan-Meier Survival Curve for the Entire Cohort



Question 3: Comparison of Survival Between Groups

Question: Compare survival curves between males and females.

```
# Convert sex to a factor for easier interpretation
lung$sex <- factor(lung$sex, levels = c(1, 2), labels =
  c("Male", "Female"))

# Fit the Kaplan-Meier survival model
km_fit <- survfit(Surv(time, status) ~ sex, data = lung)

# Plot the survival curves
ggsurvplot(km_fit, data = lung,
  title = "Kaplan-Meier Survival Curves by Sex",
  xlab = "Time (days)",
  ylab = "Survival Probability",
  conf.int = TRUE, # Show confidence intervals
  palette = c("blue", "red"), # Colors for groups
  risk.table = TRUE, # Show risk table
  pval = TRUE) # Show p-value from log-rank test
```



Answer: The survival curves indicate differences in survival probabilities between males and females.

Kaplan-Meier Survival Curves by Sex

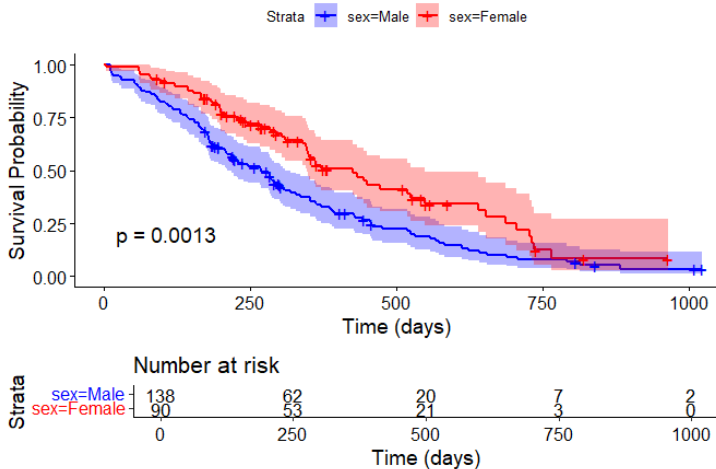


Figure: Survival Curves by Sex

Question 4: Cox Proportional Hazards Model

Question: Fit a Cox model with age and sex as covariates.

```
cox_model <- coxph(Surv(time, status) ~ sex, data = lung)
summary(cox_model)
```

Call:

```
coxph(formula = Surv(time, status) ~ sex, data = lung)
n= 228, number of events= 165
```

	coef	exp(coef)	se(coef)	z	Pr(> z)
sexFemale	-0.5310	0.5880	0.1672	-3.176	0.00149 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
sexFemale	0.588	1.701	0.4237	0.816

The results indicate that sex is statistically significant ($p = 0.0186$)

Question 5: Model Diagnostics

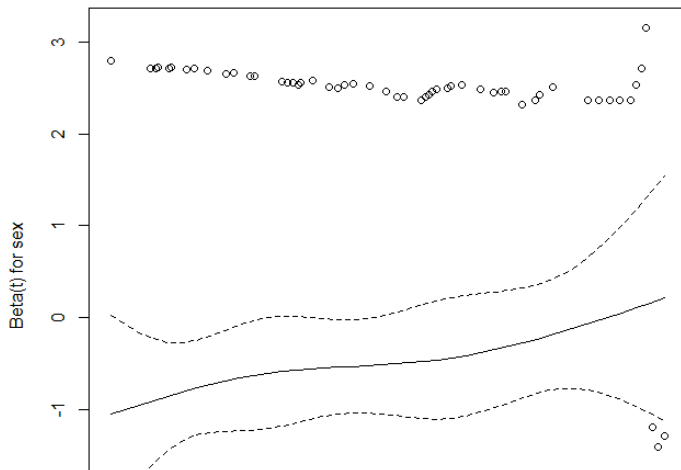
Question: Check the proportional hazards assumption.

R Code:

```
cox.zph_result <- cox.zph(cox_model)
print(cox.zph_result)
plot(cox.zph_result)
```



Answer: The results of the Schoenfeld residuals test suggest that the proportional hazards assumption holds, as no significant deviations are indicated.



Question 6: Weight Loss Analysis

Question: Fit a Cox model including weight loss.

R Code:

```
cox_model_wt_loss <- coxph(Surv(time, status) ~ wt.loss,
  data = lung)
summary(cox_model_wt_loss)
```

Answer:

Call:

```
coxph(formula = Surv(time, status) ~ wt.loss, data = lung)
```

n= 214, number of events= 152

(14 observations deleted due to missingness)

```
coef exp(coef) se(coef) z Pr(>|z|)
```

```
wt.loss 0.001319 1.001320 0.006079 0.217 0.828
```



Question 7: Subgroup Analysis

Question: Fit survival curves based on ECOG performance score.

R Code:

```
km_fit_ecog <- survfit(Surv(time, status) ~ ph.ecog, data = lung)
ggsurvplot(km_fit_ecog,
title = "Survival Curves by ECOG Performance Score",
xlab = "Days",
ylab = "Survival Probability",
conf.int = TRUE)
```



Answer: The survival curves demonstrate how survival probabilities vary with different ECOG performance scores.

Survival Curves by ECOG Performance Score

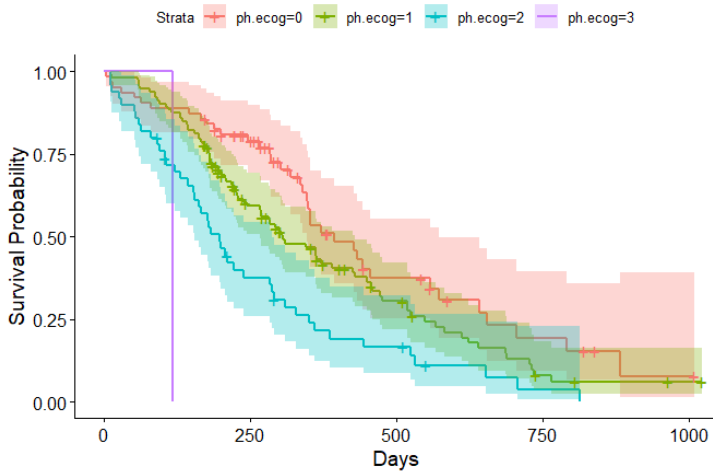


Figure: Survival Curves by ECOG Performance Score

Problem - 5



5 Problem - 5

5.1 Solution of Problem - 5



Problem - 5

The dataset simulates the results of a diagnostic test used to detect a particular disease, consisting of 1,000 individuals. It includes two primary variables: **disease_status**, a binary variable indicating whether an individual has the disease (1) or does not have the disease (0), with a disease prevalence set at 10%; and **test_results**, a binary variable representing the outcome of the diagnostic test, where a positive result is indicated by (1) and a negative result by (0). This dataset provides a foundation for analyzing the effectiveness of the diagnostic test through various epidemiological metrics.



- ① Using the provided dataset, calculate:
 - ▶ True Positives (TP)
 - ▶ True Negatives (TN)
 - ▶ False Positives (FP)
 - ▶ False Negatives (FN)
- ② Based on the confusion matrix from Question 1:
 - ▶ Compute sensitivity and specificity.
 - ▶ Interpret these values in the context of public health.
- ③ Construct the ROC curve and calculate the AUC. Discuss what this value indicates about the test's ability.



Solution of Problem - 5

```
# Load necessary libraries
```

```
library(dplyr)
```

```
library(pROC)
```

```
# Load the dataset
```

```
data <- read.csv("diagnostic_test_data.csv")
```

```
# Question 1: Confusion Matrix Calculation
```

```
TP <- sum(data$disease_status == 1 & data$test_results == 1)
```

```
TN <- sum(data$disease_status == 0 & data$test_results == 0)
```

```
FP <- sum(data$disease_status == 0 & data$test_results == 1)
```

```
FN <- sum(data$disease_status == 1 & data$test_results == 0)
```

```
# Print Confusion Matrix Elements
```

```
confusion_matrix <- data.frame(
```

```
TP = TP, TN = TN, FP = FP, FN = FN )
```

```
print(confusion_matrix)
```



R Code

Question 2: Sensitivity and Specificity

```
sensitivity <- TP / (TP + FN)
specificity <- TN / (TN + FP)
```

Print Sensitivity and Specificity

```
cat("Sensitivity:", sensitivity, "\n")
cat("Specificity:", specificity, "\n")
```

Question 3: ROC Curve Analysis

```
roc_obj <- roc(data$disease_status, data$test_results)
plot(roc_obj)
auc_value <- auc(roc_obj)
```

Print AUC

```
cat("AUC:", auc_value, "\n")
```

